

**ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**  
**ORGANISATION OF ISLAMIC COOPERATION (OIC)**  
**Department of Computer Science and Engineering (CSE)**

**QUIZ 3 SET A****SUMMER SEMESTER, 2024-2025****DURATION: 20 MINUTES****FULL MARKS: 15**

**SWE 4805: Software Verification and Validation**

Write all the answers on the question paper.

Answer **all 1 (one)** questions. Figures in the right margin indicate full marks of questions.

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**ID:**

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1.

Consider the following Java method that determines whether a given year is a leap year.

```
1 public class LeapYear {
2
3     public static boolean isLeapYear(int year) {
4
5         if (year % 400 == 0)
6             return true;
7
8         if (year % 100 == 0)
9             return false;
10
11        if (year % 4 == 0)
12            return true;
13
14        return false;
15    }
16 }
```

(CO1)  
(PO1)

The following test cases are available:

| Test Case | Input | Expected Output |
|-----------|-------|-----------------|
| T1        | 2000  | true            |
| T2        | 1900  | false           |
| T3        | 2024  | true            |
| T4        | 2023  | false           |

- a) Generate any **six** valid mutants of the given program using the mutation categories covered in class: **Value Mutation, Decision Mutation, and Statement Mutation**. For each mutant, clearly indicate the mutation category and show the modified line of code. 6
- b) For the mutants created in part (a), construct a mutation matrix indicating whether each test case **Kills (K)** or **Survives (S)** each mutant. 5
- c) Using the mutation matrix obtained in part (b), calculate the **Mutation Score** of the test suite. 4

Clearly show:

- Number of mutants generated
- Number of killed mutants
- Mutation Score formula
- Final Mutation Score (as a percentage)